Chapter 3

Monte Carlo Integration

Chapter 1 showed that Bayesian inference is integration and introduced the Monte Carlo estimator as a way around quadrature. Chapter 2 showed how to generate the draws that the estimator needs. This chapter makes Monte Carlo a reliable tool.

In this chapter, we are interested in knowing (a) how accurate an estimate is and how many draws it needs (Sections 3.2 and 3.3), (b) what to do when direct sampling is impossible or wasteful (Section 3.4), and (c) how to get more accuracy from the same number of draws (Section 3.5).

The campaign of Chapter 1 remains our running example: posterior $\mathrm{Beta}(14, 226)$ for the conversion rate θ, purchase amounts $Y \sim \mathrm{Gamma}(2, 30)$ euros, and the non-conjugate log-odds model of Example 1.4.

3.1 From expectation to integration

Example 3.1 (How often does a buyer spend more than 100 euros?)**.** The marketing team wants to know what fraction of buyers spend more than 100 euros. Under the model $Y \sim \mathrm{Gamma}(2, 30)$ of Example 1.3, the answer is an integral of the density $p(y) = \dfrac{y\, e^{-y/30}}{900}$:

$$\Pr(Y > 100) = \int_{100}^{\infty} p(y)\,\mathrm{d}y = e^{-10/3}\left(1 + \tfrac{10}{3}\right) \approx 0.1546.$$

When we write the same integral with an indicator function, $\displaystyle\int_0^{\infty} \mathbf{1}\{y > 100\}\, p(y)\,\mathrm{d}y$, it is the expectation $\mathbb{E}[\mathbf{1}\{Y > 100\}]$. The Monte Carlo recipe is therefore to simulate buyers and count those above 100 euros:

Draws N	**Estimate**	**Standard error**
100	0.1700	0.0378
1 000	0.1450	0.0111
10 000	0.1547	0.0036
100 000	0.1527	0.0011

Each estimate is within about two standard errors of the exact value; Section 3.2 explains the last column.

3.1.1 Integrals as expectations

Definition 3.1 (Expectation as an integral)**.** Let X be a random variable (or vector) with density f on a set $\mathcal{X}$, and $h : \mathcal{X} \to \mathbb{R}$ a function with $\mathbb{E}_f|h(X)| < \infty$. The expectation of $h(X)$ is

$$\mu_h = \mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x)\, f(x)\, \mathrm{d}x. \tag{3.1}$$

The Monte Carlo idea reads (3.1) from right to left: *any* integral that can be written as "function times density" is an expectation, and expectations can be estimated by averages of simulated values. Three ways to cast an integral in this form recur throughout this course.

- **Probabilities** are expectations of indicators: $\Pr(X \in A) = \mathbb{E}_f[\mathbf{1}\{X \in A\}]$.
- **Posterior summaries** are expectations under the posterior, $f = \pi(\cdot \,|\, \mathbf{y})$, as in (1.7).
- **Any integral over an interval** is an expectation under the uniform distribution: for $U \sim \mathcal{U}(a, b)$,

$$\int_a^b g(x)\, \mathrm{d}x = (b-a) \int_a^b g(x)\, \frac{1}{b-a}\, \mathrm{d}x = (b-a)\, \mathbb{E}[g(U)]. \tag{3.2}$$

For example, the Beta function $B(14, 226) = \displaystyle\int_0^1 t^{13}(1-t)^{225}\, \mathrm{d}t \approx 4.627 \times 10^{-24}$, which normalizes the campaign posterior, is estimated by (3.2) with 10^5 uniform draws as 4.678×10^{-24}, a relative standard error of 1.3%. The justification for replacing an expectation by an average is the strong LLN, already used in Proposition 1.2: if $X_1, X_2, \ldots$ are i.i.d. with density f and $\mathbb{E}_f|h(X)| < \infty$, the average $\dfrac{1}{N}\displaystyle\sum_{i=1}^{N} h(X_i)$ converges to μ_h with probability one.

3.1.2 The posterior predictive probability

Example 3.2 (Will at least 10 of the next 100 visitors buy?)**.** The retailer expects 100 more visitors tomorrow and will call the campaign a success if at least 10 of them buy. Given θ, the number of buyers $\tilde{S}$ among 100 visitors is Binomial$(100, \theta)$. We do not know θ, so we average over its posterior. The **posterior predictive probability** is

$$\Pr(\tilde{S} \geq 10 \,|\, \mathbf{y}) = \int_0^1 \Pr(\tilde{S} \geq 10 \,|\, \theta)\, \pi(\theta \,|\, \mathbf{y})\, \mathrm{d}\theta = \mathbb{E}_{\pi(\theta \,|\, \mathbf{y})}\big[q(\theta)\big],$$

$$q(\theta) = \Pr(\tilde{S} \geq 10 \,|\, \theta) = \sum_{k=10}^{100} \binom{100}{k} \theta^k (1-\theta)^{100-k}.$$

A *two-stage simulation* estimates it without computing q: draw $\theta^{(i)} \sim \mathrm{Beta}(14, 226)$, then $\tilde{S}^{(i)} \sim \mathrm{Binomial}(100, \theta^{(i)})$, and count the fraction of $\tilde{S}^{(i)} \geq 10$. With $N = 10\,000$ we obtain 0.1006 ± 0.0059; the exact value, available here through the beta-binomial distribution, is 0.1016.

The two-stage idea works for any model we can simulate forwards, even when no exact formula exists. We improve this estimator in Section 3.5.

3.2 The classical Monte Carlo estimator

Definition 3.2 (Classical Monte Carlo estimator)**.** Let $X_1, \dots, X_N \overset{\text{iid}}{\sim} f$. The **classical Monte Carlo estimator** of $\mu_h = \mathbb{E}_f[h(X)]$ is $\hat{\mu}_N = \dfrac{1}{N}\sum_{i=1}^{N} h(X_i)$.

Proposition 3.1 (Unbiasedness and variance)**.** *If* $\sigma_h^2 = \mathrm{Var}_f(h(X)) < \infty$*, then*

$$\mathbb{E}[\hat{\mu}_N] = \mu_h, \qquad \mathrm{Var}(\hat{\mu}_N) = \frac{\sigma_h^2}{N}.$$

Proof. Linearity of expectation gives $\mathbb{E}[\hat{\mu}_N] = \dfrac{1}{N}\sum_i \mathbb{E}[h(X_i)] = \mu_h$. The variables $h(X_i)$ are independent, so their variances add: $\mathrm{Var}(\hat{\mu}_N) = \dfrac{1}{N^2}\sum_i \mathrm{Var}(h(X_i)) = \sigma_h^2/N$. □

The **Monte Carlo standard error** is the standard deviation of the estimator, $\sigma_h/\sqrt{N}$. It depends on the unknown σ_h, which we estimate from the same draws by the sample standard deviation

$$\hat{\sigma}_h = \Big(\frac{1}{N-1}\sum_{i=1}^{N}\big(h(X_i) - \hat{\mu}_N\big)^2\Big)^{1/2}, \qquad \widehat{\mathrm{SE}} = \frac{\hat{\sigma}_h}{\sqrt{N}}.$$

For an indicator with success probability p, $\sigma_h^2 = p(1-p)$: for $\Pr(Y > 100)$, $\sigma_h = \sqrt{0.1546 \times 0.8454} = 0.362$ and the standard error with $N = 1000$ is 0.0114, in line with the table of Example 3.1.

Two features of Proposition 3.1 deserve emphasis: the estimator has no bias, whatever N; and its error decreases like $N^{-1/2}$ with a constant σ_h that depends on the problem, not on the dimension of X.

3.3 Convergence and error assessment

3.3.1 Confidence intervals and their coverage

The CLT of Proposition 1.2 turns the standard error into a confidence interval: for large N,

$$\hat{\mu}_N \pm 1.96\,\frac{\hat{\sigma}_h}{\sqrt{N}} \tag{3.3}$$

contains μ_h with probability close to 0.95. We can check the promise by simulation. In this case, repeating the estimation of $\Pr(Y > 100)$ with $N = 1000$ draws 2000 times, the interval (3.3) contains the exact value in 94.9% of the repetitions. Figure 3.1 shows how a single run converges: the **running mean** $\hat{\mu}_k$, $k = 1, \dots, N$, with its 95% band, which narrows like $k^{-1/2}$.

3.3.2 How many draws?

Turning (3.3) around answers a planning question: to estimate μ_h within $\pm\varepsilon$ with 95% confidence, we need

$$N \geq \left(\frac{1.96\,\sigma_h}{\varepsilon}\right)^2. \tag{3.4}$$

For $\Pr(\theta > 0.05 \mid \mathbf{y}) = 0.6897$, $\sigma_h = \sqrt{0.6897 \times 0.3103} = 0.463$, and the required sample sizes are 8 222 for $\varepsilon = 0.01$, 32 887 for $\varepsilon = 0.005$, and 822 166 for $\varepsilon = 0.001$. Each extra digit costs a factor 100. In practice σ_h is unknown: a **pilot run** of 1 000 draws gives $\hat{\sigma}_h = 0.467$ and the plan $N \approx 33\,500$ for $\varepsilon = 0.005$, close to the exact requirement.

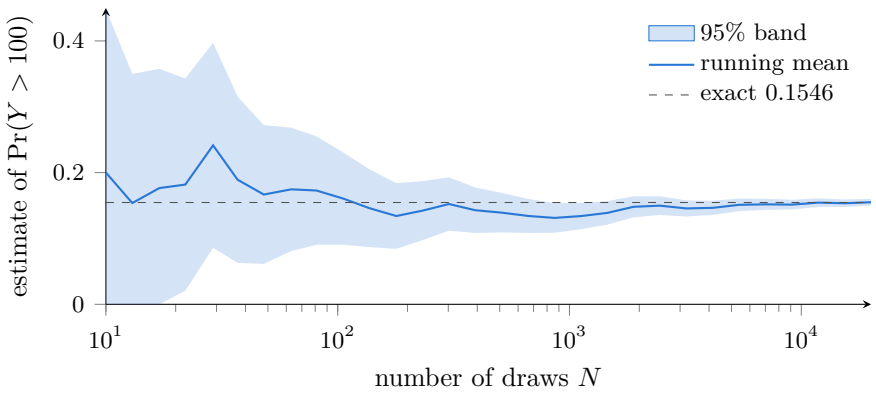


Figure 3.1: Running Monte Carlo estimate of $\Pr(Y > 100)$ with its 95% confidence band (3.3), on a logarithmic horizontal axis. The band narrows like $N^{-1/2}$ and contains the exact value 0.1546.

3.3.3 When the CLT fails

Proposition 3.1 and (3.3) require a finite variance. Real business quantities do not always have one.

Example 3.3 (Customer lifetime value)**.** The lifetime value of a customer (i.e., the total spending over the relationship) is typically heavy-tailed: most customers spend little, a few "whales" spend enormous amounts. A common model is the Pareto distribution with minimum $x_m = 10$ euros and tail index $\alpha = 1.5$, $\Pr(X > x) = (x_m/x)^\alpha$ for $x \geq x_m$, simulated by inverse transform as $X = x_m(1-U)^{-1/\alpha}$. Its mean $\alpha x_m/(\alpha-1) = 30$ euros is finite, but

$$\mathbb{E}[X^2] = \int_{x_m}^{\infty} x^2 \, \frac{\alpha x_m^\alpha}{x^{\alpha+1}} \, \mathrm{d}x = \infty \qquad \text{for } \alpha \leq 2,$$

so, the variance is infinite. The LLN still applies, but the CLT does not. Figure 3.2 shows the running mean of one million draws: it drifts upward and jumps each time a large value arrives, and after 1 000 draws it is 24.7 (the largest draw so far: 588 euros), after 100 000 draws 29.8 (largest: 36 269 euros). Across 1 000 repetitions with $N = 1000$, the estimates have a median 28.5, a 90% range [25.6, 37.3], and a maximum of 158. In other words, they are skewed and occasionally wild, and the sample standard deviation, hence the reported standard error, changes erratically from run to run.

Remark 3.1 (Diagnostics)**.** Always look at the running mean, not just the final number. Sudden jumps, a band that does not shrink like $N^{-1/2}$, or estimates that change markedly between seeds are signs of heavy tails. Remedies include transforming the quantity, for example, the median or a log scale, or importance sampling.

3.4 Importance sampling

3.4.1 A rare event: purchases above 400 euros

The finance team wants the probability of a single purchase above 400 euros to size a fraud check.

Exactly, $\Pr(Y > 400) = e^{-40/3}(1 + 40/3) \approx 2.32 \times 10^{-5}$: one buyer in about 43 000. With $N = 10\,000$ plain Monte Carlo draws, we expect 0.23 purchases above 400 euros; in 80% of repetitions, the estimate is exactly zero, and the relative RMSE is 2.15 (an error of 215%).

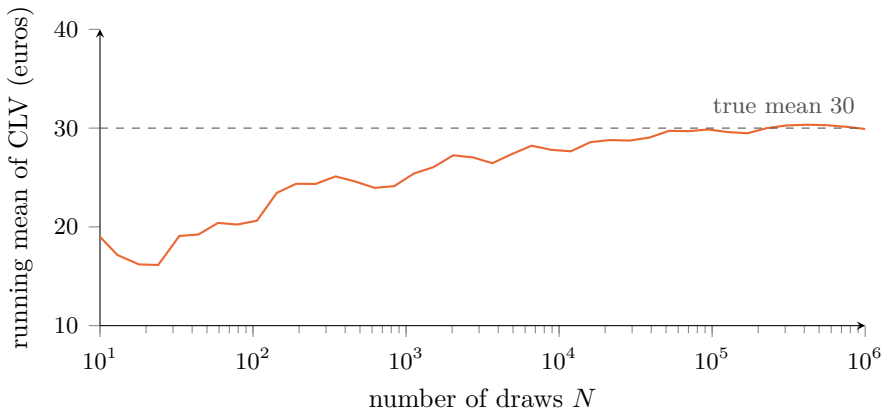


Figure 3.2: Running mean of one million Pareto customer lifetime values ($x_m = 10$, $\alpha = 1.5$, true mean 30). With infinite variance, convergence is slow and irregular.

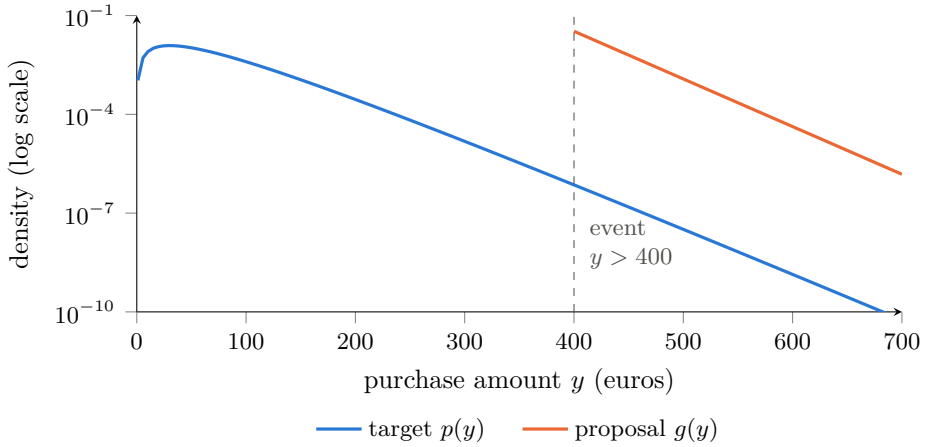


Figure 3.3: Target density $p(y)$ of purchase amounts and importance proposal $g(y)$, on a logarithmic vertical axis. The proposal puts all its mass on the rare event $y > 400$.

The problem is not the estimator but where the draws land: almost none land where the event happens.

Suppose instead that we draw from a **proposal** density g that lives on the event, for instance 400 plus an exponential with mean 30 (Figure 3.3).

Every draw now satisfies $y > 400$, but it is too likely under g compared with the target p. We correct by weighting each draw by the ratio $w(y) = p(y)/g(y)$. With $N = 10\,000$ proposal draws,

$$\widehat{\Pr}(Y > 400) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{y_i > 400\} \frac{p(y_i)}{g(y_i)} = 2.3224 \times 10^{-5} \pm 3.2 \times 10^{-8},$$

a relative error of 0.04%; over 200 repetitions the relative RMSE is 0.07%, three thousand times smaller than with plain Monte Carlo.

3.4.2 The importance sampling identity

Proposition 3.2 (Importance sampling)**.** *Let f be the target density and g a proposal density with $g(x) > 0$ wherever $h(x)f(x) \neq 0$. Then*

$$\mathbb{E}_f[h(X)] = \int h(x)\,\frac{f(x)}{g(x)}\,g(x)\,\mathrm{d}x = \mathbb{E}_g\big[h(X)\,w(X)\big], \qquad w(x) = \frac{f(x)}{g(x)}.$$

*With $X_1, \dots, X_N \overset{iid}{\sim} g$, the **importance sampling estimator** $\hat{\mu}_N^{\mathrm{IS}} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} h(X_i)\,w(X_i)$ is unbiased, and its variance is $\dfrac{1}{N}\operatorname{Var}_g(h(X)w(X))$, finite if and only if $\mathbb{E}_g[h^2w^2] = \displaystyle\int h^2 f^2/g < \infty$.*

The **importance weights** $w(X_i)$ correct for sampling from the wrong distribution. The proof is the displayed identity together with Proposition 3.1 applied to hw under g. Which proposal is best?

Proposition 3.3 (Optimal proposal)**.** *The variance of $\hat{\mu}_N^{\mathrm{IS}}$ is minimized by $g^\star(x) = |h(x)|f(x)/\displaystyle\int |h|f$. If $h \geq 0$, this proposal has zero variance.*

Proof. By the Cauchy–Schwarz inequality, $\big(\int |h|f\big)^2 = \big(\int \frac{|h|f}{\sqrt{g}}\sqrt{g}\big)^2 \leq \int \frac{h^2f^2}{g}\int g = \mathbb{E}_g[h^2w^2]$, with equality when $g \propto |h|f$. The variance is $\mathbb{E}_g[h^2w^2] - \mu_h^2$, so it is minimal at $g^\star$; when $h \geq 0$, $\displaystyle\int |h|f = \mu_h$ and the variance is zero. □

$g^\star$ is not usable, since it requires the unknown μ_h, but it tells us what a good proposal looks like: *proportional to* $|h|f$, that is, concentrated where both the integrand and the density are large. For the rare event, $|h|f$ is the density restricted to $y > 400$, and our shifted exponential is close to it. A second practical rule follows from the finite-variance condition: the proposal should have *heavier tails* than $|h|f$, so that the weights stay bounded.

Unnormalized targets. In Bayesian computation the target is known only up to a constant, $f = \tilde{f}/Z$ with $Z = \displaystyle\int \tilde{f}$ (for a posterior, $\tilde{f} = p(\mathbf{y} \mid \theta)\pi(\theta)$ and $Z = p(\mathbf{y})$). Using unnormalized weights $\tilde{w}(x) = \tilde{f}(x)/g(x)$, the identity of Proposition 3.2 gives two estimators.

Definition 3.3 (Evidence estimate and self-normalized importance sampling)**.** With $X_1, \dots, X_N \overset{\text{iid}}{\sim} g$ and $\tilde{w}_i = \tilde{f}(X_i)/g(X_i)$:

$$\hat{Z}_N = \frac{1}{N}\sum_{i=1}^{N} \tilde{w}_i, \qquad \hat{\mu}_N^{\mathrm{SN}} = \frac{\sum_{i=1}^N \tilde{w}_i\,h(X_i)}{\sum_{i=1}^N \tilde{w}_i} = \sum_{i=1}^{N} \bar{w}_i\,h(X_i), \quad \bar{w}_i = \frac{\tilde{w}_i}{\sum_j \tilde{w}_j}.$$

The **effective sample size (ESS)** is $\mathrm{ESS} = \big(\sum_i \tilde{w}_i\big)^2 / \sum_i \tilde{w}_i^2 = 1/\sum_i \bar{w}_i^2$.

$\hat{Z}_N$ is unbiased for Z: it is Proposition 3.2 with $h = 1$ and f replaced by $\tilde{f}$.

The **self-normalized** estimator $\hat{\mu}_N^{\mathrm{SN}}$ is a ratio of two unbiased estimators. It has a small bias of order $1/N$ but converges to μ_h with probability one by the LLN, and it needs no normalizing constant.

The ESS lies between 1 (i.e., one weight dominates) and N (i.e., all weights equal), and it

approximates the number of independent draws from the target that would give the same precision.

An ESS that is a small fraction of N signals a poor proposal. As in Remark 1.3, weights are computed on the log scale and normalized with the log-sum-exp trick.

3.4.3 The non-conjugate posterior, revisited

Example 3.4 (Importance sampling for the log-odds model)**.** Return to Example 1.4, with unnormalized posterior $\tilde{\pi}(\eta \mid \mathbf{y}) = L(\eta)\,\mathcal{N}(\eta; \mu_0, \sigma_0^2)$. Its mode is $\hat{\eta} = -2.803$ (conversion rate 0.0572) and the curvature of $\log \tilde{\pi}$ at the mode corresponds to a standard deviation of 0.260. A **Laplace proposal** is a distribution centered at the mode with this scale; we use a Student-t with 4 degrees of freedom for heavier tails. We compare it with two other proposals, each with $N = 10\,000$ draws; the reference values from quadrature are $p(\mathbf{y}) = 9.528 \times 10^{-21}$, $\mathbb{E}[\theta \mid \mathbf{y}] = 0.0576$ and $\Pr(\theta > 0.05 \mid \mathbf{y}) = 0.6845$.

Proposal g	$\hat{p}(\mathbf{y})$	$\mathbb{E}[\theta \mid \mathbf{y}]$	$\Pr(\theta > 0.05 \mid \mathbf{y})$	ESS
Laplace, Student-t_4	9.57×10^{-21}	0.0579	0.693	9 455
prior $\mathcal{N}(\mu_0, 0.5^2)$	9.59×10^{-21}	0.0576	0.688	6 629
too narrow $\mathcal{N}(\hat{\eta}, 0.05^2)$	6.03×10^{-21}	0.0568	0.807	100

Over 200 repetitions with $N = 2000$, the relative RMSE of $\hat{p}(\mathbf{y})$ is 0.5% for the Laplace proposal, 1.6% for the prior, and 54% for the narrow proposal. The narrow proposal almost never visits the tails of the posterior, where its weights would be huge; its estimates are too low in most runs, and they are unbiased only on average over rare runs with enormous weights. Its ESS of 100 out of 10 000 gives the warning without knowing the answer. Compared with accept–reject (Section 2.5.1), importance sampling uses every draw, also estimates the evidence, and does not need a bound M.

The same diagnostic explains the uniform draws of Example 1.6. Estimating $\int e^{-\|\theta\|^2/2}\,\mathrm{d}\theta = (2\pi)^{d/2}$ with 10^5 draws:

d	**Uniform on $[-5,5]^d$: estimate/truth**	$\mathcal{N}(0, 2^2 I_d)$: estimate/truth	ESS
1	1.001	1.001	66 215
5	1.017	1.004	12 614
10	0.561	0.946	1 790
20	0.003	0.926	38

A Gaussian proposal that roughly matches the target is far better than a uniform one, but even so the ESS falls from 66% of the draws in one dimension to 0.04% in twenty: the weights degenerate as the dimension grows, because a mismatch of a factor 2 in scale compounds across dimensions. This is the main limitation of importance sampling, and the reason why Chapters 5 and 6 turn to Markov chains.

3.5 Variance reduction techniques

3.5.1 A noisy estimate

The two-stage estimator of Example 3.2 has a standard deviation of 0.0094 with $N = 1000$ posterior draws, almost 10% of the value 0.1016 being estimated. To halve the error, we would need four times more draws. The techniques of this section reduce the variance instead by exploiting the structure that plain Monte Carlo ignores. We apply four of them to the

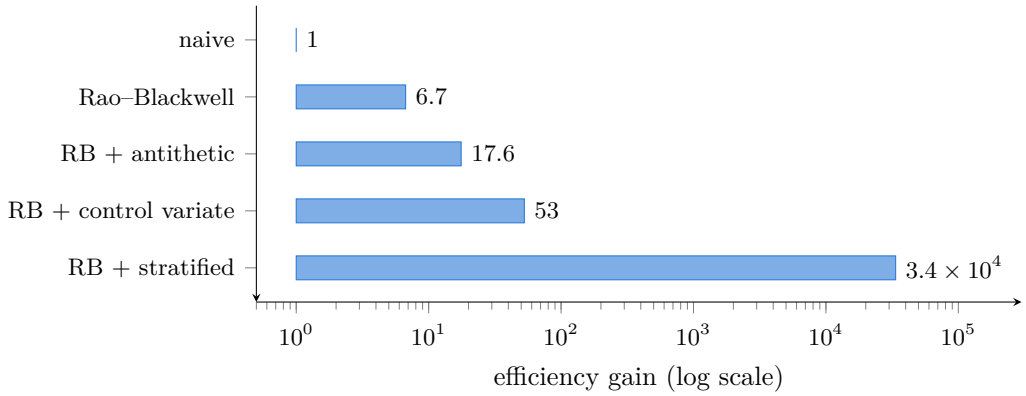


Figure 3.4: Efficiency gains for the predictive probability $\Pr(\tilde{S} \geq 10 \,|\, \mathbf{y})$ with $N = 1000$ posterior draws (500 repetitions), on a logarithmic scale. RB: Rao–Blackwellized.

predictive probability; all use the same $N = 1000$ posterior draws, produced by inverse transform $\theta = F^{-1}(U)$ with F the Beta$(14, 226)$ CDF.

Figure 3.4 summarizes the gains, measured by the **efficiency gain**. The variance of the naive estimator divided by the variance of the improved one, that is, the factor by which the number of draws would have to grow to reach the same precision without the technique.

estimator	mean of estimates	standard deviation	efficiency gain
naive two-stage	0.10200	0.00944	1.0
Rao–Blackwell	0.10208	0.00366	6.7
Rao–Blackwell + antithetic	0.10149	0.00225	17.6
Rao–Blackwell + control variate	0.10154	0.00130	52.8
Rao–Blackwell + stratified	0.10157	0.00005	33 622

The exact value is 0.10157. We now explain each technique.

3.5.2 Four techniques

Rao–Blackwellization. The naive estimator averages $\mathbf{1}\{\tilde{S} \geq 10\}$, which is random twice: through θ and through $\tilde{S}$ given θ. The second source of randomness is unnecessary, because the conditional expectation $q(\theta) = \Pr(\tilde{S} \geq 10 \,|\, \theta)$ is a binomial tail probability that we can compute exactly.

Proposition 3.4 (Rao–Blackwell)**.** *Let (X, Y) be random and $q(X) = \mathbb{E}[h(X, Y) \,|\, X]$. Then $\mathbb{E}[q(X)] = \mathbb{E}[h(X, Y)]$ and $\mathrm{Var}(q(X)) \leq \mathrm{Var}(h(X, Y))$.*

Proof. The first identity is the law of iterated expectations. The **law of total variance** gives $\mathrm{Var}(h) = \mathbb{E}[\mathrm{Var}(h \,|\, X)] + \mathrm{Var}(\mathbb{E}[h \,|\, X]) = \mathbb{E}[\mathrm{Var}(h \,|\, X)] + \mathrm{Var}(q(X))$, and the first term is non-negative. □

Averaging $q(\theta^{(i)})$ instead of indicators removes the binomial noise $\mathbb{E}[\mathrm{Var}(h \,|\, \theta)]$ and gives a gain of 6.7.

Antithetic variables. Draw $N/2$ uniforms U_i and use each twice, as $\theta = F^{-1}(U_i)$ and $\theta' = F^{-1}(1 - U_i)$. Both are posterior draws, but they are negatively correlated: a small U_i gives a small θ and a large θ'.

Proposition 3.5 (Antithetic variables)**.** *Let $Y = h(F^{-1}(U))$ and $Y' = h(F^{-1}(1-U))$ with $\mathrm{Var}(Y) = \sigma^2$ and correlation $\rho = \mathrm{Corr}(Y, Y')$. The average of $N/2$ independent pairs $(Y+Y')/2$ is unbiased with variance $\frac{\sigma^2}{N}(1+\rho)$. If h is monotone, $\rho \le 0$, so the variance is at most that of N independent draws.*

Proof. $\mathrm{Var}(\frac{Y+Y'}{2}) = \frac{\sigma^2}{2}(1+\rho)$, and averaging $N/2$ independent pairs divides by $N/2$. For monotone h, Y and Y' are monotone functions of U in opposite directions, hence negatively correlated. □

Here q is increasing in θ, and the antithetic pairs add a factor of about 2.6 on top of Rao–Blackwellization.

Control variates. We know the posterior mean of θ exactly, $\mathbb{E}[\theta \mid \mathbf{y}] = 14/240$, and θ is strongly correlated with $q(\theta)$ (correlation 0.931). If a sample has too many large θ's, it probably also overestimates $\mathbb{E}[q(\theta)]$; we correct for this.

Proposition 3.6 (Control variates)**.** *Let $Y = h(X)$ and let $C = c(X)$ have known mean $\mathbb{E}[C]$. For any constant b, $\hat{\mu}_b = \frac{1}{N}\sum_i (Y_i - b\,(C_i - \mathbb{E}[C]))$ is unbiased. Its variance is minimized at $b^\star = \mathrm{Cov}(Y, C)/\mathrm{Var}(C)$, where it equals $\frac{\sigma_Y^2}{N}(1-\rho_{YC}^2)$, with ρ_{YC} the correlation of Y and C.*

Proof. $\mathrm{Var}(Y - bC) = \sigma_Y^2 - 2b\,\mathrm{Cov}(Y, C) + b^2\,\mathrm{Var}(C)$ is a quadratic in b, minimized at $b^\star$ with minimum $\sigma_Y^2 - \mathrm{Cov}(Y, C)^2/\mathrm{Var}(C) = \sigma_Y^2(1-\rho_{YC}^2)$. □

In practice, $b^\star$ is estimated from the same draws. With $\rho = 0.931$ the predicted gain is $1/(1-0.931^2) \approx 7.5$ on top of Rao–Blackwellization: $6.7 \times 7.5 \approx 50$, close to the observed 52.8.

Stratification. Plain uniform draws leave gaps and clusters (Figure 3.5). Stratification splits $(0, 1)$ into K intervals of equal length, the **strata**, and draws the same number of points in each.

Proposition 3.7 (Stratified sampling)**.** *Split $(0, 1)$ into K strata of length $1/K$ and draw N/K uniforms in each. The average of $h(F^{-1}(U))$ over all draws is unbiased, with variance $\frac{1}{NK}\sum_{k=1}^{K}\sigma_k^2 \le \frac{\sigma^2}{N}$, where σ_k^2 is the variance of $h(F^{-1}(U))$ for U uniform in stratum k.*

The inequality holds because the total variance σ^2 is the average within-stratum variance plus the variance between stratum means; stratification removes the second part.

With $K = N = 1000$ strata of one draw each, the within-stratum variation of the smooth function $q(F^{-1}(u))$ is tiny, and the gain reaches 33 622. The method then resembles a quadrature rule, and like one, it suffers from the curse of dimensionality: in d dimensions, K strata per axis require K^d cells.

3.5.3 The probability of beating the benchmark

Example 3.5 (Variance reduction for $\Pr(\theta > 0.05 \mid \mathbf{y})$)**.** The indicator $h(\theta) = \mathbf{1}\{\theta > 0.05\}$ is an increasing function of $\theta = F^{-1}(U)$, so antithetic and stratified sampling apply (Rao–Blackwellization and control variates offer nothing obvious here). With $N = 1000$ draws and 500 repetitions, the standard deviations of the estimates are 0.0135 (plain), 0.0106 (antithetic) and 0.00048 (stratified with 1000 strata). The antithetic gain is modest, because a jump function is only weakly correlated with its mirror image; stratification is dramatic, because only the one stratum containing $F(0.05)$ has any variation at all.

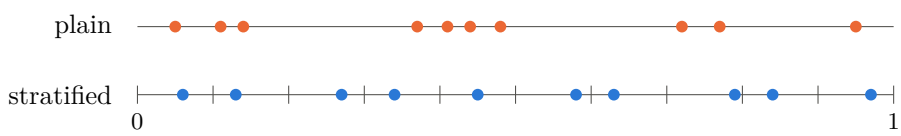


Figure 3.5: Ten uniform draws: plain sampling leaves gaps and clusters; stratified sampling places exactly one draw in each tenth of the interval.

Summary

Takeaways

- Any integral of the form "function times density" is an expectation, and Monte Carlo estimates it by an average. Probabilities, posterior summaries and predictive probabilities all fit this form.
- The classical estimator is unbiased with standard error $\sigma_h/\sqrt{N}$, estimated from the draws; the interval $\hat{\mu} \pm 1.96\,\widehat{\mathrm{SE}}$ has close to 95% coverage when $\sigma_h < \infty$. Plan sample sizes with $N \geq (1.96\,\sigma_h/\varepsilon)^2$, using a pilot run.
- Heavy tails (infinite variance) break the standard error: inspect running means.
- Importance sampling draws from a proposal g and reweights by f/g. It handles rare events, unnormalized posteriors (self-normalized estimator) and the evidence. A good proposal resembles $|h|f$ with heavier tails; the effective sample size diagnoses bad ones, and weights degenerate in high dimension.
- Rao–Blackwellization, antithetic variables, control variates and stratification reduce the variance by exploiting known conditional expectations, symmetry, known means, and even coverage of the uniform input.

Further reading

Chapters 3 and 4 of Robert and Casella (2004) treat Monte Carlo integration, importance sampling, and convergence monitoring in depth, and Chapter 3 of Gamerman and Lopes (2006) covers the same material with a Bayesian focus. Importance sampling for posterior computation, including weight diagnostics, is discussed in Chapter 10 of Gelman et al. (2013), and Chapter 11 of Bishop (2006) presents sampling methods from a machine-learning perspective.